

Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

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Abstract

Climate-driven shifts in stock productivity reach fishing fleets unevenly when species portfolios differ across fleet segments and the sector-level allocation of catch rights cannot adjust to changes in relative productivity. We quantify this distributional channel for the multi-species small pelagic fishery of Central-South Chile (anchoveta, sardine, jack mackerel), managed under sector-specific quota regimes with a fixed industrial–artisanal allocation. Our framework couples a Bayesian state-space model of stock dynamics, fit to 2000–2024 official assessment data, with a vessel-level trip equation; climate enters through an indirect biomass channel and a direct weather channel. Under a six-model CMIP6 ensemble spanning two emissions scenarios and two time windows, projected fleet-level effort falls by 13–15 percent for the artisanal fleet but only 0.8–1.0 percent for the industrial fleet — an asymmetry of roughly fifteen to one driven by differential portfolio exposure and weather sensitivity, with portfolio exposure decomposed between fleets and within the artisanal fleet by vessel type. The findings imply that climate-adaptation policy in Chile should revisit the sector-level allocation of catch rights, allowing it to respond to relative productivity shifts rather than remain fixed independent of the climate state.

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1 Introduction

Climate change is reshaping the productivity of marine fisheries, but the incidence of this reshaping across fleet segments is rarely uniform. Quota-allocation regimes that codify the relative shares of industrial and artisanal (small-scale) fleets were typically designed under stationary-climate assumptions, so a fleet-asymmetric productivity shock acts as an unscripted reallocation of harvest rights. Whether such regimes should anticipate this reallocation depends on the projected magnitude and asymmetry of fleet-level effort responses under credible climate scenarios—a quantity the empirical literature has yet to deliver for most major multi-species fisheries.

We provide that quantity for the small pelagic fishery (SPF) of Central-South Chile (Regions V–X and XIV), part of the country’s largest fishery—approximately 86% of national fish landings by industrial and artisanal sectors in 2019 (SUBPESCA, 2020)—and the segment in which all three target species, anchoveta (*Engraulis ringens*), sardina común (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*), coexist in non-negligible shares of both fleet portfolios. These three species share habitat and are coupled through trophic and market linkages (Alheit & Niquen, 2004; Arancibia et al., 2019; Dresdner et al., 2013). The fishery is managed through a sector-specific quota architecture: a fixed industrial-versus-artisanal allocation splits the Total Allowable Catch between the two segments, while within-sector distribution operates through Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) for the industrial fleet and the *Régimen Artesanal de Extracción* (RAE) for the artisanal fleet, with limited cross-sector transferability. Because the two segments target different species portfolios, a stock-asymmetric climate shock translates directly into a fleet-asymmetric effort response. Cross-country evidence indicates that Chile is projected to lose part of its maximum catch potential under climate change: Cheung et al. (2010) classify Chile among a group of countries with moderate projected declines (6–13% by 2055 under SRES A1B), including the contiguous United States, China and Brazil; the available climate-fisheries

economics literature is patchy globally and concentrated in a few well-studied regions (Sumaila et al., 2011). To our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.¹

Our bioeconomic framework combines two empirical components. First, a Bayesian state-space model of multi-species stock dynamics — fit to 2000–2024 official IFOP and SPRFMO assessments — estimates, for each species, two semi-elasticities of productivity with respect to sea-surface temperature and log-chlorophyll-a anomalies. We call these parameters *climate shifters*. Because they enter the biological law of motion as coefficients on exogenous climate forcings, they remain interpretable when applied to climate states outside the 2000–2024 sample. Second, a vessel-level negative binomial trip equation — estimated separately for the industrial and artisanal fleets, with year fixed effects that absorb aggregate non-climate shocks (notably the 2019–2020 social and pandemic disruptions) — maps prices, allocated harvest, weather, and regulatory closures into annual fishing effort. Climate reaches fleet-level effort through two channels: an *indirect biomass channel* via the climate shifters, and a *direct weather channel* via vessel-specific exposure to severe winds. We evaluate the framework against a six-model CMIP6 ensemble under SSP2-4.5 and SSP5-8.5, at mid-century (2041–2060) and end-of-century (2081–2100). Throughout, we report two sources of uncertainty separately: dispersion across climate models (which future climate trajectory we project against) and dispersion in the Bayesian posterior of the structural parameters (uncertainty about the elasticities themselves)

We obtain three sets of results. First, the distributional incidence of climate-driven productivity shocks across fleets is sharply asymmetric: the artisanal fleet contracts by 13.2–15.0% across CMIP6 scenarios, while the industrial fleet contracts by only 0.8–1.0% — a ratio of roughly fifteen to one. Two economic mechanisms compound to produce this asymmetry. Each fleet is exposed to climate-driven productivity shifts through a distinct species portfolio, and the fixed industrial–artisanal allocation binds, preventing the cross-sector reassignment of harvest rights that would otherwise equalize marginal returns under the new relative productivities. The result instantiates, in a climate-shock setting, the portfolio-and-access framework of Kasperski & Holland (2013) and Cline et al. (2017): segments with narrower species portfolios bear sharper variance, and the wel-

¹The only empirical study of Chilean fisher behavior that we are aware of is Peña-Torres et al. (2017), which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

fare cost of that exposure rises when institutional access constraints prevent rebalancing (Oken et al., 2021). Because adaptation responses at the individual fisher level are themselves heterogeneous and context-dependent (Jardine et al., 2020; Zhang & Smith, 2011), the policy reading is direct: cross-sector transferability is the first-order rent-reallocation margin for attenuating disproportionate fleet-level losses, and any reform—the size, direction, and shadow price of admissible transfers—is quantifiable from the posterior we report. Second, the long-run change in intrinsic productivity is large and negative for anchoveta and sardine—the two coastal-upwelling stocks that dominate the artisanal portfolio—with floor effects for sardine under SSP5-8.5 end-of-century. Third, the climate shifters are sharply identified for these two stocks but not for jack mackerel, the species that anchors the industrial portfolio. We document this null with four converging tests (spatial-domain robustness across nested coastal aggregations, dual-source biomass with the Northern Chilean acoustic series, an OROP-PS coherence diagnostic, and basin-scale ENSO Niño 3.4 as an alternative shifter) and treat the jack mackerel productivity factor as fixed at its prior mean throughout the projections.

The remainder of the paper is organized as follows. Section 2 describes the economic and institutional setting of the Chilean SPF. Section 3 presents the data, the stock-dynamics model, the trip equation, and the CMIP6 projection design (Section 3.4). Section 4 reports the identified climate shifters, the comparative-statics projections of stock productivity, and the projections of fleet-level effort (Section 4.3). Section 5 discusses policy implications and limitations; Section 6 concludes.

2 The small pelagic fishery in Chile

The Chilean small pelagic fishery (SPF) is the largest fishery in the country, accounting for approximately 86% of national fish landings by the industrial and artisanal sectors in 2019 (SUBPESCA, 2020). The fishery is divided into two latitudinal zones with distinct species compositions. The northern zone (Regions I–IV) is dominated by anchoveta (*Engraulis ringens*) and jack mackerel (*Trachurus murphyi*), while sardine (*Strangomera bentincki*) is effectively absent. In contrast, the Central-South zone (Regions V–X and XIV) includes all three species in non-negligible shares within both artisanal and industrial landings. Because this region supports a genuinely multi-

species fishery operating under a fixed sector-level allocation regime, it provides the relevant setting for evaluating the distributional consequences of climate-induced shifts in stock productivity, and is therefore the focus of this paper. The principal ports servicing this zone include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. The three species are harvested primarily by purse-seine fleets that differ in vessel size, operating range, and species composition.

The historical evolution of the fishery illustrates the portfolio mechanism developed in the remainder of the paper. The jack mackerel fishery was initially concentrated in northern Chile but shifted toward the Central-South zone during the mid-1980s (Peña-Torres et al., 2017), where fishing grounds have traditionally operated within 50 nautical miles of the coast. Between 1987 and 2004, approximately 85% of jack mackerel landings were directed to the reduction industry (Peña-Torres et al., 2017), reflecting the importance of fishmeal and fish-oil export markets in determining the value of the industrial portfolio. Sardine and anchoveta, by contrast, are jointly harvested using the same gear and fishing grounds. The two species share habitat and fishing technology, making species differentiation at the point of capture effectively impossible (Dresdner et al., 2013). As a result, substitution between them is constrained primarily by quota regulations rather than by biological or technological limitations.

The status of the three stocks has evolved substantially over time. Central-South anchoveta was classified as collapsed through 2018, reclassified as overexploited in 2019, and has been harvested within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within MSY reference levels, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel remained overexploited until 2018 and has since been harvested within MSY limits.

The Chilean SPF is managed through an annual Total Allowable Catch (TAC; *Cuota Global*) that is divided between the industrial and artisanal sectors, with small fractions reserved for research and contingency purposes. TACs are further subdivided across regions and seasons, and unused quotas may be reassigned within the fishing year. Anchoveta and sardine are managed under a mixed-species quota system: although each species has an independent quota, substitution between them is permitted within the regulatory framework.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP). Class A

licenses were allocated according to historical catch shares, whereas Class B licenses—representing up to 15% of the industrial allocation—have been distributed through sealed-bid first-price auctions since 2015. Peña-Torres et al. (2022) document that these auctions have exhibited low participation, limited capacity to reflect economies of scale, and indications of coordinated bidding behavior. The artisanal sector operates under a regulated open-access regime, with regional TACs distributed through the *Régimen Artesanal de Extracción* (RAE) according to area, vessel size, landing site, or fisher organization.

Biological closures (*vedas*) constitute a central management instrument for sardine and anchoveta. In the Central-South zone, reproductive closures extend from July to October in Regions IX–XIV and from August to October in Regions V–VIII, while recruitment closures occur from January through February across all regions. In total, these regulations generate 151 closed fishing days per year in Regions V–VIII and 182 days in Regions IX–XIV. Jack mackerel is not subject to seasonal biological closures.

Beyond the within-sector quota architecture, the partition of each TAC between sectors is fixed by statute. Article~47 of the General Law of Fisheries and Aquaculture (LGPA) governed this de jure partition during the estimation window (2018–2024); a 2025 reform, Ley_N~21.752, fixes a new partition in force until 2040 (Table~1). Under Article~47, the artisanal sector held 78% of the Central-South anchoveta and sardina com'un TACs and 10% of the jurel TAC; Ley~21.752 raises these to 90% for both small pelagics and to 30% in V–Los~R'ios and 15% in Los~Lagos for jurel. The de jure partition admits a single inter-sectoral re-allocation channel under Article₅₅T of the LGPA: industrial titulares and artisanal armadores may cede assigned tonnage across the sectoral boundary subject to ex-ante SUBPESCA authorisation. The channel is heavily used: SERNAPESCA quota control reports for 2013–2024 document that the industrial sector cedes a stable 85–99% of its assigned anchoveta and sardina com'un TACs to artisanal cessionaries every year (Online Appendix H.3), while jurel cessions are an order of magnitude smaller and shift direction across years. Our empirical specifications are therefore identified on realised landings, which are post-cession; the operational allocation reflects the joint Article₄₇–Article₅₅~T regime, not the de jure fractioning alone. Within sectors, transferability differs sharply: industrial coefficients trade under the LTP regime introduced in 2013, while artisanal RAE assignments remain locked

to the regional registry. We return to the welfare implications of the 2025 redistribution under climate stress in Section~5 and Online Appendix H.

Table 1: Statutory sectoral partition of the Central-South small pelagic TACs, pre- and post-Ley 21.752 (2025).

Stock	TAC 2024 (kt)	Artisanal share of TAC (%)	
		Art. 47 LGPA	Ley 21.752
Anchoveta (V–Los Lagos)	211.3	78	90
Sardina común (V–Los Lagos)	290.6	78	90
Jurel (V–Los Ríos)	578.6	10	30
Jurel (Los Ríos–Los Lagos)	80.6	10	15

Notes. TAC 2024 is the Cuota Global de Captura fixed by SUBPESCA for the calendar year, in thousand tonnes. The pre-2025 fractioning regime follows the Article 47 schedule of the LGPA (DS 430/1991 of the Ministerio de Economía, Fomento y Reconstrucción, consolidated text of Law 18.892/1989) [SERNAPESCA2024-cuota]. The 2025 regime corresponds to Article 1 of Ley N° 21.752 [Ley21752-2025; Boletin17096-21], in force until 31 December 2040. Items 6 and 7 of Article 1 assign 90% of the anchoveta and sardina común TACs to the artisanal sector throughout the V–Los Lagos macrozone (single block). Items 4 and 5 partition jurel into two sub-macrozones: 30% artisanal in V–Los Ríos and 15% artisanal in Los Lagos. The SERNAPESCA quota report aggregates Los Ríos and Los Lagos into a single jurel quota of 80.6 kt; the row in the table applies the Los Lagos share (15%) as a conservative approximation, with the understanding that the true sub-block share lies between 15% and 30% depending on the Los Ríos proportion of the aggregate quota. A use-it-or-lose-it adjustment rule applies in the northern macrozone and is omitted from the table because the present paper focuses on the Central-South. Industrial-to-artisanal inter-sectoral transfers are subject under Article 6 to a specific patent of 1.3 monthly tax units per transferred ton, with a minimum of 250 monthly tax units; artisanal armadores are exempt from the patent on transfers in either direction.

3 Methods

Our empirical strategy targets two structural objects—climate shifters of stock productivity and fleet-specific elasticities of fishing effort—and combines them to compute counterfactual fleet re-

sponses under future climate paths. First, we identify the climate shifters: species-specific semi-elasticities of intrinsic stock productivity with respect to sea-surface temperature and chlorophyll-a anomalies. We estimate them within a Bayesian state-space model of multi-species stock dynamics, with a Schaefer law of motion and priors taken from official IFOP and SPRFMO assessments. Second, we estimate fleet-specific elasticities of annual fishing effort with respect to allocated harvest, ex-vessel prices, weather exposure, and regulatory closures, using a negative binomial count model fitted separately for the artisanal and industrial fleets, with year fixed effects that absorb aggregate non-climate shocks. Third, we compute long-run comparative statics under a six-model CMIP6 ensemble (SSP2-4.5 and SSP5-8.5, mid- and end-of-century), propagating the climate shifters through the stock-dynamics steady state to long-run productivity and through the trip equation to fleet-level effort.

3.1 Historical data

We use data requested from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) covering the 2013–2024 period. The dataset includes trip-level microdata with detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and gear type, ports of departure and landing, fishery codes, haul timing and location, species composition, retained catch, and trip activity. We also requested annual information on stock abundance and vessel landings by port, county, region, country, and species. Finally, we use ex-vessel prices reported monthly or annually by port, county, region, country, and species. These prices reflect payments from processing plants to fishers at the point of first sale and are obtained through IFOP’s landing surveys, which do not necessarily cover all market transactions.

Environmental covariates are obtained from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Salinity, sea surface temperature, and current speed and direction come from the Global Ocean Physics Reanalysis (GLORYS12V1), which provides data at $1/12^\circ$ horizontal resolution with 50 vertical levels ([E.U. Copernicus Marine Service Information, 2025c](#)). Surface wind speed and direction are taken from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, at 0.125° horizontal resolution and hourly frequency ([E.U. Copernicus Marine Service Information, 2025b](#)).

Chlorophyll-a concentrations come from the Global Ocean Colour dataset at \$ \$4 km horizontal resolution (E.U. Copernicus Marine Service Information, 2025a). All environmental data were retrieved daily (hourly for winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 41°S (Figure 1).

Sample structure for jack mackerel biomass identification The jack mackerel biomass sample over the 2000–2024 estimation window is informative for identification in 18 of 25 years. Sixteen years contribute a point estimate from the official IFOP hydroacoustic assessment, and two additional years (2012, 2015) enter the likelihood as left-censored observations at the assessment’s lower detection limit. The remaining seven years carry no biomass observation; their latent state is identified dynamically by the law of motion of Section 3.3.1, together with the priors on $(r^0, K, \sigma_{\text{proc}}, \sigma_{\text{obs}})$ from official IFOP and SPRFMO assessments and on (ρ^{SST}, ρ^{CHL}) from the reduced-form stress tests of Online Appendix A. This reduced informational content is one of the load-bearing facts behind the non-identification result for the jack mackerel climate shifter (Section 4.1).

Two external series—jack mackerel biomass from the Northern acoustic zone (Arica–Iquique–Antofagasta, available since 2010) and the SPRFMO transzonal spawning biomass index for the Southeast Pacific—were considered as auxiliary information. The Northern series correlates with the Centro-Sur (CS) series at $r = 0.82$ over $N = 7$ overlapping years, while the SPRFMO index correlates with CS at $r = 0.11$ over $N = 17$ overlapping years. We treat these numbers as suggestive rather than conclusive: a seven-point Pearson correlation is at best descriptive, but the contrast in sign and magnitude is consistent with a Centro-Sur stock unit that shares short-run dynamics with the Northern Chilean assessment and is only weakly coupled to the SPRFMO-scale aggregate. Neither external series enters the structural model’s likelihood—as a prior or as an observation—to avoid introducing information about coastal climate forcing through the indirect route of Northern or transzonal biomass.

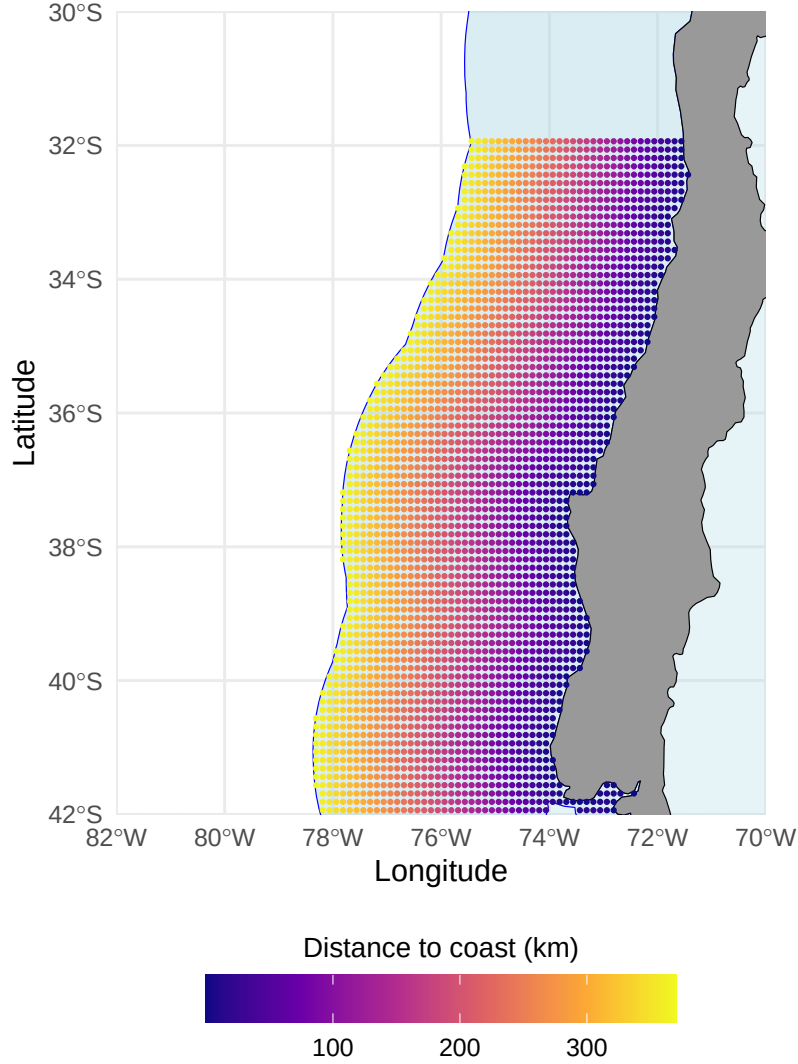


Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

3.2 Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive using a six-model ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR), selected to span the CMIP6 equilibrium climate sensitivity distribution. We use monthly outputs for two future scenarios: SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). Section 3.4 describes the projection horizons, baseline period, and delta-method procedure used to translate CMIP6 outputs into our empirical models. The per-model decomposition of projection uncertainty appears in Online Appendix F for stock-level intrinsic

productivity and in Online Appendix G for fleet-level trips.

3.3 Econometric models

3.3.1 Stock dynamics

Our identification target in this subsection is the vector of climate semi-elasticities $(\rho_i^{SST}, \rho_i^{CHL})_{i=1}^3$ —the structural parameters that translate sea-surface temperature and chlorophyll-a anomalies into shifts in long-run stock productivity. We recover them by inverting a Bayesian state-space model of multi-species stock dynamics in which a Schaefer transition with a log-linear climate shifter on the intrinsic growth rate governs the latent biomass $B_{i,t}$ of stock $i \in \{\text{anchoveta, sardina común, jack mackerel}\}$. The Schaefer form (the $\theta = 1$ case of the Pella–Tomlinson family) is adopted on identifiability grounds: with $N = 25$ annual observations per stock, the shape parameter θ is notoriously weakly identified, and imposing $\theta = 1$ is a standard identification device in the small-pelagic literature (Hilborn & Walters, 1992; Maunder & Punt, 2013). The law of motion is

$$B_{i,t+1} = B_{i,t} + r_{i,t} B_{i,t} \left(1 - \frac{B_{i,t}}{K_i}\right) - C_{i,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_{\text{proc},i}^2), \quad (1)$$

where $C_{i,t}$ is the observed annual catch, K_i is the stock-specific carrying capacity, and $\sigma_{\text{proc},i}$ is the standard deviation of the stochastic recruitment innovation. Climate enters the dynamics through a log-linear shifter on the intrinsic growth rate,

$$r_{i,t} = r_i^0 \exp\left(\rho_i^{SST} (SST_{t-1} - \overline{SST}) + \rho_i^{CHL} (\log CHL_{t-1} - \overline{\log CHL})\right), \quad (2)$$

so ρ_i^{SST} and ρ_i^{CHL} are the semi-elasticities of intrinsic stock productivity to a one-degree Celsius SST anomaly and to a one-log-point chlorophyll-a anomaly, respectively. Because these parameters enter the law of motion as coefficients on exogenous climate forcings, they are invariant to the joint historical distribution of (SST, B, C) —transportable, in the structural sense, to climate regimes outside the 2000–2024 estimation sample. This transportability matters because the comparative-statics exercise of Section 4.3 evaluates the climate shifter at projected SST anomalies that lie

several historical standard deviations beyond the estimation envelope.

Latent biomass is linked to the official survey-based estimates of spawning stock biomass $B_{i,t}^{\text{obs}}$ —published by IFOP for anchoveta and sardina común in Centro-Sur Chile and by SPRFMO for jack mackerel—through a log-normal measurement equation,

$$\log B_{i,t}^{\text{obs}} = \log B_{i,t} + u_{i,t}, \quad u_{i,t} \sim \mathcal{N}(0, \sigma_{\text{obs},i}^2). \quad (3)$$

Environmental conditions are summarized annually by sea surface temperature (SST) and chlorophyll-a concentration (CHL), averaged over the Centro-Sur region within the Chilean EEZ and lagged one year to respect the pre-determined ordering of environmental forcing relative to year- t biomass. The choice of these two covariates follows a large biological literature that identifies thermal regime and primary productivity as the dominant local drivers of small-pelagic dynamics in the Humboldt Current System (Axbard, 2016; Cahuin et al., 2009; Yáñez et al., 2014; see Cheung et al., 2008; Jennings et al., 2008).

We estimate the state-space specification by Hamiltonian Monte Carlo in Stan. With $N = 25$ annual observations per stock, joint identification of $(r_i^0, K_i, \rho_i^{\text{SST}}, \rho_i^{\text{CHL}})$ requires the prior information embedded in the IFOP and SPRFMO single-species assessments; Online Appendix A documents the resulting prior elicitation protocol and the boundary problem under maximum likelihood. We estimate three nested specifications and adopt the climate-augmented version of Eq. (2) as our primary object of interpretation, with the nested predictive comparison relegated to Online Appendix B and convergence diagnostics ($\hat{R} < 1.01$ and bulk/tail ESS > 400 for all top-level parameters) to Online Appendix D. The seven jurel Centro-Sur years without a survey estimate enter the likelihood as latent biomass states identified dynamically; two further years (2012, 2015) enter through a log-normal censored-observation term at the assessment detection limit, with all associated uncertainty propagated directly into the posterior of $(\rho_{\text{jurel}}^{\text{SST}}, \rho_{\text{jurel}}^{\text{CHL}})$.

The state-space formulation in Eq. (1)–(3) separates process from observation noise, enforces a pre-determined ordering between year- $(t - 1)$ environmental forcing and year- t latent biomass, and yields structural climate semi-elasticities that are transportable to climate regimes outside the 2000–2024 estimation window—the property that supports the comparative-statics exercise of

Section 4.3 at projected SST anomalies several historical standard deviations beyond the estimation envelope.

Figure 2 shows that anchoveta, sardina común, and jack mackerel exhibit distinct biomass trajectories over the 2000–2024 window, with no evidence of strong interannual co-movement across species once scale differences are accounted for. Harvest pressure is visibly associated with these fluctuations; jack mackerel in particular experienced an abrupt decline during the late 2000s, consistent with the combined effects of intense fishing and unfavorable environmental conditions.

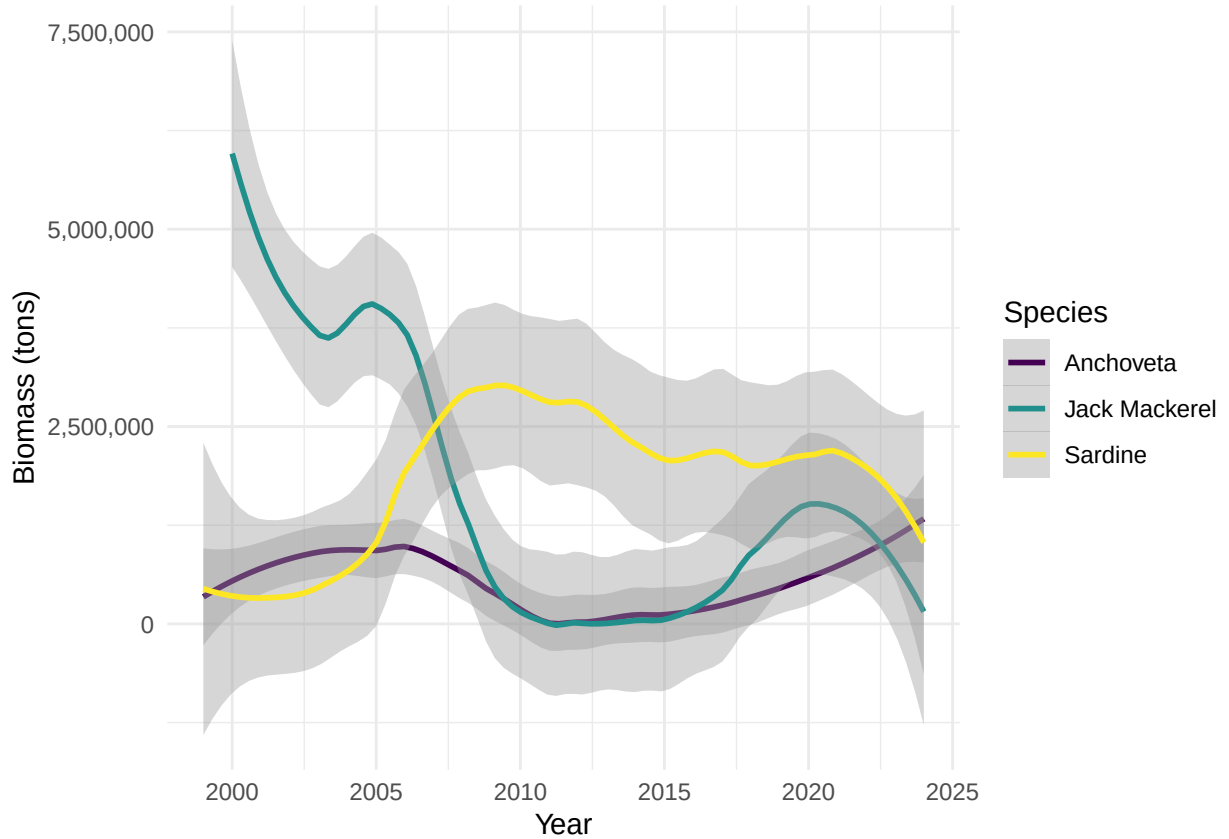


Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024)

3.3.2 Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (2015). Because the available logbook data record only purse-seine operations, the unit of observation is a vessel–year. We estimate separate effort models for the industrial and arti-

sanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology. A complementary approach, used in Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery, models daily participation and target-species choice through a discrete choice framework with species distribution models as proxies for daily availability. The annual NB specification we adopt here is the natural counterpart in a setting where vessel-year is the operational unit at which TACs are allocated under the LTP, and where daily logbook coverage is too sparse to support a daily DCM for the Centro-Sur Chilean SPF over the 2013–2024 window.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \quad \lambda_{vy} = \exp(U'_{vy}\beta), \quad (4)$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y . Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction for both fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy}, H_{vy}^{alloc}, Z_v, O_{vy}]. \quad (5)$$

Output prices p_{sy} are species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP’s manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Following Dresdner et al. (2013), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the model aims to characterize vessels’ effort responses to changes in prices, quotas,

and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual effort. Cross-validation against the SERNAPESCA vessel-level landings database (transparency request AH010T0006857; 2013–2024 Centro-Sur) confirms that the resulting panel is representative of the purse-seine artisanal segment. For the industrial fleet, the panel covers 40 of 59 registered vessels (68% by count) and aggregate landings within -10% of the SERNAPESCA all-gear total, with portfolio-weighted catch composition matching SERNAPESCA aggregates within two percentage points across the three target species. For the artisanal fleet, the panel covers 859 of 2,689 registered vessels (32% by count) but accounts for approximately 70% of total Centro-Sur artisanal landings of the three species, reflecting the concentration of catch in the larger purse-seine vessels for which logbook submission is mandatory. Smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside the LTP quota architecture and are not modeled in the trip equation; they appear in the SERNAPESCA aggregate landings used as the likelihood input for the biomass state-space model.

Quota shares do not enter Eq. (4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω_{vs}^r denote vessel v 's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}_{sy}^r$ for species s in year y and region r —obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H_{vy,s}^{alloc} = \omega_{vs}^r \bar{Q}_{sy}^r, \quad H_{vy}^{alloc} = \sum_s H_{vy,s}^{alloc}. \quad (6)$$

Each vessel is assigned to its administrative region based on its modal departure port in the logbook records. Artisanal TACs are assigned at the regional level (regions V, VIII, IX, XIV, X), while industrial TACs follow the regulatory zone structure established by SUBPESCA (zones V–IX and XIV–X for jack mackerel; zone V–X for sardine and anchoveta). This regional construction introduces cross-sectional variation in H_{vy}^{alloc} that reflects the heterogeneous quota environments

faced by vessels operating in different parts of the Centro-Sur fishery.

Following Kasperski (2015), the trip equation enters allocated harvest *by species* rather than as a scalar aggregate: the regressor vector contains $H_{vy,s}^{alloc}$ for $s \in \{\text{anchoveta, sardine, jack mackerel}\}$ with separate semi-elasticities β_h^s . Ex-vessel prices likewise enter at $(year, region)$ resolution, mapped to each vessel through its modal departure region. For the artisanal fleet, the response to allocated anchoveta is allowed to vary by vessel type through an interaction $H_{vy,anchoveta}^{alloc} \times \text{TIPO_EMB}$, motivated by the observation in the data that small-vessel categories (LM = *lancha motor*, the dominant 77% of the artisanal panel) exhibit a near-zero elasticity to anchoveta quota, while medium-vessel categories (BM = *bote motor*, 8%) exhibit a strong positive elasticity in the spirit of the Kasperski (2015) benchmark. The interaction is estimated only for TIPO_EMB categories with at least 50 vessel-year observations (BM, LM, UNK); three small categories with $N < 50$ (BR, BRV, L) collectively represent less than 0.1% of artisanal effort and quota and are reported with an aggregated rather than category-specific coefficient. The Kasperski-aligned specification is preferred by an information criterion of $\Delta\text{AIC} = -167.9$ for the artisanal fleet (and -5.0 for the industrial fleet) relative to the legacy scalar specification, with multicollinearity diagnostics (VIFs below 2.5 across all six regressors and both fleets) confirming separate identification of the three β_h^s .

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector O_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel-year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of adverse-weather days is computed as the count of days per year in which wind speed exceeds 8 m/s at the environmental grid point nearest to the vessel's COG.² Second,

²The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. We selected this threshold via AIC comparison across 8, 10, and 12 m/s cut-offs on the trip-equation fit.

the number of days closed for biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel’s COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in regions V–VIII; July–October in regions IX–XIV) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. The purpose of Eq. (4), however, is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices, quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

3.4 Projection approach

We assess the impact of climate change on the Chilean small pelagic fishery through two channels: (i) a *direct weather channel*, in which projected changes in wind speed alter the number of bad-weather days that constrain fishing operations and thus enter the negative binomial trip equation; and (ii) an *indirect biomass channel*, in which projected changes in SST and CHL modify each stock’s intrinsic productivity $r_{i,t}$ through the climate shifter $(\rho_i^{SST}, \rho_i^{CHL})$ of Eq. (2) and, in turn, the law of motion in Eq. (1).

Climate projections are derived from a six-model CMIP6 ensemble—IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, and MPI-ESM1-2-HR—under two Shared Socioeconomic Pathways: SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions). The ensemble was selected to span the CMIP6 distribution of equilibrium climate sensitivity. We apply the delta method (Burke et al., 2015; Free et al., 2019) independently to each model, which preserves the observed interannual variability and fine-scale spatial structure from satellite records while imposing the climate-change signal from CMIP6. For each model and variable, the delta is computed as the difference (additive for SST and wind speed) or, for chlorophyll-a, the difference of natural logarithms (so that the resulting $\Delta \log \text{CHL}$ enters the shifter on its native estimation scale), be-

tween the CMIP6 future climatology and a spliced historical baseline that combines the CMIP6 historical run for 2000–2014 with the SSP2-4.5 run for 2015–2024 (CMIP6 historical runs end in 2014, so the splice is required to match the 2000–2024 estimation window of the structural model). For CESM2, which lacks chlorophyll under SSP2-4.5 in the NCAR CMIP6 catalogue, we splice the historical 2000–2014 segment with the SSP5-8.5 2015–2024 segment for chlorophyll only; the two scenarios are virtually indistinguishable in the near term and the substitution preserves CESM2’s contribution to SSP5-8.5 chlorophyll deltas. We consider two future time windows: mid-century (2041–2060) and end-of-century (2081–2100). Comparative statics are then reported integrating over both the structural-parameter posterior and the cross-model dimension of the ensemble; the variance decomposition of the resulting projection uncertainty is reported in Online Appendix F for stock-level intrinsic productivity and in Online Appendix G for fleet-level trips.

For the direct weather channel, monthly wind speed deltas from CMIP6 are applied additively to the daily wind speed observations at each vessel’s center of gravity (COG) over 2013–2024, and bad-weather days (wind speed exceeding 8 m/s) are recounted vessel by vessel under each scenario; the resulting Δdays_{bw} enters the NB trip equation through the estimated semi-elasticity β_{weather} , retaining vessel-level heterogeneity in baseline weather exposure. Although the cross-model median CMIP6 wind anomaly is modest (below +0.5 m/s under SSP5-8.5 end-of-century), the historical daily wind distribution at the COG sits sufficiently close to the 8 m/s threshold that the cross-vessel median Δdays_{bw} rises by +10 days under SSP2-4.5 mid-century and +23.5 days under SSP5-8.5 end-of-century. Note that *uas* and *vas* are missing from the CESM2 archive in the CMIP6 DRS for this period, so the direct channel is computed from the five remaining ensemble members (CNRM-ESM2-1, GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, UKESM1-0-LL) while the indirect biomass channel retains the full six-member ensemble where SST and chlorophyll deltas are available. For the indirect biomass channel, we evaluate the climate shifter of Eq. (2) at the posterior draws of $(\rho_i^{SST}, \rho_i^{CHL})$ from the full state-space specification, taking ΔSST and $\Delta \log CHL$ as the scenario- and window-specific CMIP6 anomalies relative to the 2000–2024 estimation baseline. This yields a full posterior distribution over $r_{i,t}^*/r_i^0$ under each climate regime—a long-run comparative-statics object, in the sense that it describes the steady-state productivity of each stock under a counterfactual climate, not a year-by-year forecast of biomass. Forward

simulation of biomass and catch trajectories under endogenous harvest rules is left to future work; the posterior reported here is archived to facilitate that extension.

Our framework is positioned as long-run comparative statics, not a year-by-year forecast, in line with the bioeconomic-projection tradition of Kasperski (2015). Because the empirical models are estimated on interannual variability rather than decadal trends, the projections capture the mechanical effect of changed environmental and weather conditions on stock productivity and fleet-level effort, while holding prices, quotas, and adaptive behaviour by fishers and managers fixed at historical levels (Auffhammer, 2018). They provide a benchmark for the direction and magnitude of climate impacts under a fixed-institution baseline; relaxations that endogenize price adjustment, quota revision, or vessel-level adaptation are signposted as natural extensions of this posterior.

4 Results

4.1 Identification of climate shifters

The central econometric object of this paper is the vector of stock-specific *climate shifters* $\{\rho_i^{SST}, \rho_i^{CHL}\}_{i=1}^3$, which modulate the intrinsic growth rate of each stock as

$$r_{i,t} = r_i^0 \exp(\rho_i^{SST} (SST_{t-1} - \overline{SST}) + \rho_i^{CHL} (\log CHL_{t-1} - \overline{\log CHL})).$$

These shifters admit a clean economic reading: ρ_i^{SST} is the semi-elasticity of the stock's intrinsic productivity r_i with respect to a sea-surface temperature anomaly of one degree Celsius, and similarly ρ_i^{CHL} is the semi-elasticity with respect to a log-point of chlorophyll-a anomaly (a common proxy for primary productivity at the base of the food web). They are the structural parameters that couple the biological law of motion $B_{i,t+1} = B_{i,t} + g_i(B_{i,t}, X_t) - C_{i,t} + \varepsilon_{i,t}$ to the physical climate, and are therefore the quantities that must be identified for the long-run projections of Section 4.3 to have meaning as *comparative statics under climate change* rather than as out-of-sample forecasts.

We adopt weakly informative priors on $\rho_i^{SST}, \rho_i^{CHL}$ derived from independent reduced-form stress tests (Appendix A) and let the 2000–2024 biomass series update them. Table 2 reports the posterior

means, 90% credible intervals, and the posterior-to-prior standard deviation ratio, which measures how much the data refine each parameter relative to the prior.

Table 2: Posterior identification of climate shifters.

Shifter	Stock	Prior $\mathcal{N}(\mu, \sigma)$	Posterior mean	Posterior 90% CI	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	$\hat{R}$
ρ^{SST}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-1.06	[-1.79, -0.38]	0.43	1.000
ρ^{SST}	Sardina comun CS	$\mathcal{N}(-2.0, 1.0)$	-2.75	[-3.46, -2.02]	0.44	1.000
ρ^{SST}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	0.42	[-1.25, 2.09]	1.01	1.000
ρ^{CHL}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-3.64	[-5.01, -2.25]	0.83	1.000
ρ^{CHL}	Sardina comun CS	$\mathcal{N}(2.1, 1.0)$	2.17	[0.96, 3.37]	0.74	1.000
ρ^{CHL}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	-0.06	[-1.72, 1.60]	1.01	1.001
ρ^{ENSO}	Jurel CS	$\mathcal{N}(0.0, 0.5)$	-0.02	[-0.81, 0.80]	0.98	1.000

Note:

Shifters are the semi-elasticities of the intrinsic growth rate r_i with respect to SST anomalies (in °C), log-CHL anomalies, and the basin-scale ENSO Niño 3.4 index, respectively. The ENSO row reports the lag-1 refit of Online Appendix E.4, in which jack mackerel productivity is forced by the basin-scale index in place of local coastal shifters; anchoveta and sardine are not refit and inherit their main-specification posteriors. Priors on coastal shifters are adopted from the stock-specific stress-test results reported in Appendix A; the ENSO prior $\mathcal{N}(0, 0.5)$ is deliberately tighter (see Online Appendix E.4). The posterior-to-prior standard-deviation ratio measures prior-to-posterior updating: values near one indicate the data carry little information, whereas values well below one indicate identification from the likelihood. All $\hat{R}$ $\leq$ 1.01.

The pattern of identification is itself informative and organises the economic interpretation of the paper. Three facts stand out.

First, for the two coastal upwelling stocks—anchoveta and sardina común—both climate shifters are identified with substantial precision. The posterior 90% credible intervals exclude zero for all four parameters, and the standard deviation ratios fall between 0.43 and 0.83, indicating that the 2000–2024 biomass series refines the priors materially. For anchoveta, a one-degree positive SST anomaly is associated with a semi-elasticity of $\rho_{\text{anch}}^{SST} = -1.06$ (CI [-1.79, -0.38]); for sardina común, the corresponding semi-elasticity is more than twice as large, $\rho_{\text{sard}}^{SST} = -2.75$ (CI [-3.46, -2.02]). Evaluated at one historical standard deviation of the SST anomaly series (estimation-sample $\hat{\sigma}_{SST} = 0.26^\circ\text{C}$ over 2000–2024), a thermal shock of that magnitude lowers the intrinsic growth rate of anchoveta by approximately 24% ($1 - e^{-1.06 \cdot 0.26}$) and that of sardina by approximately 51% ($1 - e^{-2.75 \cdot 0.26}$) relative to the climatological mean. Under a policy-relevant warming signal of $+1^\circ\text{C}$, comparable to mid-century CMIP6 SSP2-4.5 projections for the Chilean coastal zone, the implied declines are 65% for anchoveta and 94% for sardina, although $+1^\circ\text{C}$ lies approximately four historical standard deviations outside the estimation window and these pro-

jections therefore rely on the log-linear extrapolation embedded in the shifter function. Either way, the magnitudes are economically first-order for a fishery whose Centro-Sur TAC is in the low hundred-thousand-tonne range. The narrow historical SST variance ($\hat{\sigma}_{SST} = 0.26^\circ\text{C}$) also explains, mechanically, why the one-step-ahead forecasting test of Appendix B fails to discriminate between the climate-augmented and the climate-free specifications: within the estimation window the annual climate signal is small relative to the stochastic process innovation σ_{proc} , yet accumulates to economically consequential levels only over decadal horizons.

Second, the two coastal stocks respond to chlorophyll-a with **opposite signs**: $\rho_{\text{anch}}^{CHL} = -3.64$ (CI $[-5.01, -2.25]$) versus $\rho_{\text{sard}}^{CHL} = +2.17$ (CI $[+0.96, +3.37]$). At one historical standard deviation of $\log CHL$ ($\hat{\sigma}_{\log CHL} = 0.095$, a primary-productivity anomaly of roughly 10%), a positive CHL shock lowers anchoveta growth by approximately 29% while raising sardina growth by approximately 23%. This is a reduced-form footprint of the well-documented trophic asymmetry between the two species, summarised here for the non-specialist reader. Chlorophyll-a is a satellite-observable proxy for phytoplankton biomass, the primary producers at the base of the marine food web. Pulses of high chlorophyll-a indicate periods of strong primary productivity, typically driven by coastal upwelling that brings nutrient-rich deep water to the surface. Sardina común feeds directly on small phytoplankton and benefits from these pulses, so its productivity rises with chlorophyll-a anomalies. Anchoveta, by contrast, feeds on larger zooplankton whose populations are themselves preyed upon by small phytoplankton-feeders; when the food web is dominated by small primary producers, the zooplankton biomass that anchoveta relies on is suppressed, and anchoveta productivity falls. The opposite-signed semi-elasticities of ρ_{anch}^{CHL} and ρ_{sard}^{CHL} identified in Table 2 are consistent with this trophic asymmetry and are not a statistical artefact. The economic implication is first-order for this paper: the artisanal fleet, which lands both species, is not simply “climate-exposed”; it is exposed to **within-target heterogeneity** in the sign of climate effects. A warming-and-greening regime therefore does not translate into a uniform shock on the artisanal sector, but into a substitution across species that the current quota-allocation rules, indexed by species, cannot fully absorb.

Third, and in marked contrast, **neither SST nor CHL shifters are identified for jack mackerel**. The posterior standard deviations σ_{post} are indistinguishable from the prior scale ($\sigma_{\text{post}}/\sigma_{\text{prior}} \approx 1$), the posterior means lie near zero, and the credible intervals straddle zero in

both directions. This is the expected result under the null hypothesis that jack mackerel productivity at the Centro-Sur scale is not locally coupled to the Chilean coastal upwelling regime—which is biologically sensible given its transboundary Southeast Pacific stock structure managed through SPRFMO. Econometrically, the model correctly reports non-identification rather than producing a spurious coefficient, which is a strength of the Bayesian state-space approach with structurally informed priors: the data can either move the prior or, as here, confirm that they are silent about it. This is the econometric dividend of Cowles-style structural identification in fishery climate-impact assessment: the structural posterior reports the absence of local climate signal where a reduced-form benchmark would have produced an artefactually precise but structurally meaningless coefficient, which would then propagate into climate projections as if it were information.

Online Appendix E reports five complementary robustness tests of this null—three nested coastal aggregations, a dual-source extension with a Northern acoustic series, basin-scale ENSO shifters at lags 1 and 2, a joint specification with all three shifters active simultaneously, and an attempted SPRFMO regional assessment integration. All five converge on posterior-to-prior ratios between 0.94 and 1.03, indistinguishable from unity, indicating that the local non-identification is structural rather than an artefact of spatial aggregation, sample size, mis-specified forcing scale, or joint-specification convention. The relevant climate forcing for the Centro-Sur jurel stock operates at margins (behaviour, location, phenology) not captured in the elasticity of annual aggregate productivity, consistent with its transboundary SPRFMO management (Arcos et al. (2001); Peña-Torres et al. (2017)).

Figure 3 visualises the posterior–prior updating as a forest plot.

4.1.1 Identification versus short-run prediction

The identified shifters are structural parameters rather than reduced-form correlations: they enter the law of motion as coefficients on exogenous climate inputs, which licenses their use as projection inputs under climate regimes outside the 2000–2024 estimation sample. Their economic value lies in **long-run comparative statics**: given a projected climate regime X_t^* drawn from the CMIP6 ensemble, the intrinsic productivity under that regime is $r_i^* = r_i^0 \exp(\rho_i^{SST} \Delta SST_t + \rho_i^{CHL} \Delta \log CHL_t)$, and steady-state biomass, harvest, and fleet-level effort follow mechanically

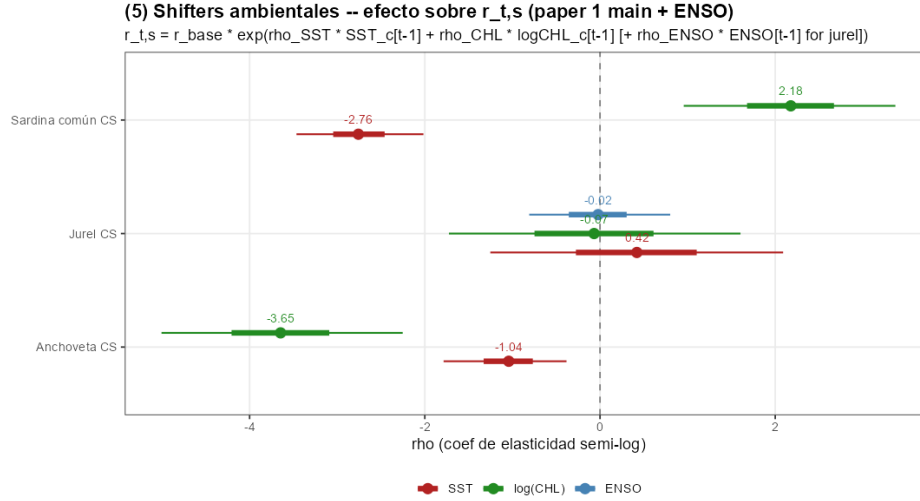


Figure 3: Posterior and prior distributions of the climate shifters $\rho_i^{SST}, \rho_i^{CHL}$ by stock. Anchoveta and sardina común show substantial posterior updating with narrow credible intervals excluding zero; jack mackerel posteriors coincide with the weakly informative prior centred at zero.

(Section 4.3). Identification of ρ does not imply superior short-run prediction: a leave-future-out cross-validation at the one-year horizon delivers $\Delta\widehat{\text{ELPD}}_{\text{LFO}} \approx 0.4$ over a climate-free baseline (Online Appendix B), as expected when process noise dominates annual climate innovations. The PSIS-LOO comparison in Online Appendix B confirms the in-sample preference for the climate-augmented specification ($\Delta\widehat{\text{ELPD}}_{\text{LOO}} = 14.59$, $\text{SE} = 4.93$); the paper relies on identification-through-fit for the economic interpretation.

4.1.2 Posterior-predictive adequacy

A stock-by-stock comparison of the smoothed latent biomass $B_{i,t}$ against the observed SSB series (Online Appendix C, Figure C.1) shows that the identified model reproduces the historical trajectory within a median 90% posterior band width of approximately 20% of the mean. Residuals are roughly Gaussian and contain no detectable autocorrelation at one-year lag (Online Appendix C, Figure C.2). Posterior-predictive diagnostics are reported in Online Appendix C; Markov-chain convergence diagnostics for all top-level parameters are reported in Online Appendix D.

4.2 Total annual trips

Table 3 reports the negative binomial estimates separately for each fleet segment. The specification includes year fixed effects, which absorb aggregate shocks affecting both fleets contemporaneously—most notably the 2019 social outbreak (*estallido social*) and the 2020–2022 COVID-19 period—and identify the structural coefficients β_H (allocated harvest) and β_{weather} (bad-weather days) from within-year cross-vessel variation. Without year fixed effects, $\hat{\beta}_{\text{weather}}^{\text{ART}}$ is approximately twice as large in magnitude, reflecting the contemporaneous correlation between high-wind years and the social/pandemic shocks. The year-FE specification is therefore the basis for the climate projections in Section 4.3; the relative trip projection $T_v^{\text{FUT}}/T_v^{\text{HIST}}$ does not depend on the year-FE values themselves, which cancel in the ratio.

The coefficients on species-specific allocated harvest reveal the heterogeneity in fleet response that the legacy scalar specification masked. For the industrial fleet, the sardine coefficient is positive and highly significant ($p < 0.001$); the anchoveta coefficient is negative but not statistically distinguishable from zero; and the jack mackerel coefficient is positive and marginally significant ($p \approx 0.05$). For the artisanal fleet, sardine remains the dominant identified channel ($\hat{\beta}_h^{\text{sard}} = 6.65 \times 10^{-4}$, $p < 0.001$). The anchoveta response is heterogeneous across vessel types: in the reference category BM (bote motor, 8% of the panel) the coefficient is large and positive ($\hat{\beta}_h^{\text{anch,BM}} = +0.0164$, $p < 0.001$), consistent with the classical Kasperski (2015) interpretation that more quota implies more trips; in LM (lancha motor, 77% of the panel) the coefficient collapses to near zero through the interaction term and ceases to be economically meaningful, indicating that the dominant artisanal segment is structurally driven by sardine rather than anchoveta. The legacy specification, which constrains the artisanal anchoveta response to a single scalar, recovers a small negative coefficient that is a weighted average of these two regimes biased toward zero by the LM weight; an information criterion comparison favours the interacted specification by $\Delta\text{AIC} = -167.9$. These coefficients provide the linkage between the trip equation and the comparative-statics projections of Section 4.3: changes in regional TACs translate into vessel-level effort adjustments through the species-specific quota-share mechanism, with the artisanal response decomposed by vessel type.

The two fleets differ markedly in price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet’s primary orientation

Table 3: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4201$)

	<i>Dependent variable:</i>	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m ³)	0.097 (0.221)	0.132*** (0.046)
Allocated harvest anchoveta ($H_{vy,anch}^{alloc}$, tons)	−0.0003 (0.0002)	0.016*** (0.003)
Allocated harvest sardine ($H_{vy,sard}^{alloc}$, tons)	0.0001*** (0.00003)	0.001*** (0.00004)
Allocated harvest jack mackerel ($H_{vy,jur}^{alloc}$, tons)	0.00002* (0.00001)	0.001 (0.001)
Price anchoveta (1000s \$/ton)	−0.004 (0.004)	0.011*** (0.003)
Price sardine (1000s \$/ton)	−0.016*** (0.003)	0.003** (0.002)
Price jack mackerel (1000s \$/ton)	0.008*** (0.003)	0.007*** (0.002)
Bad weather days	−0.0003 (0.001)	−0.001 (0.0005)
Veda days	−0.079*** (0.005)	0.013*** (0.002)
Vessel type (UNK)	−1.450*** (0.163)	0.160 (0.099)
Vessel type (BR)		−0.329*** (0.079)
Vessel type (BRV)		−0.774*** (0.079)
Vessel type (L)		−2.212*** (0.377)
Vessel type (LM)		0.289*** (0.098)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (BR)}$		−0.021*** (0.008)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (BRV)}$		−66.546*** (13.119)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (L)}$		501.903 (501.942)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (LM)}$		−0.017*** (0.003)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (UNK)}$		−0.013*** (0.003)
Constant	15.763*** (2.605)	−3.697*** (0.519)
Year fixed effects	Yes	Yes
Pseudo- R^2 (Cragg-Uhler)	0.302	0.631
Observations	319	4,201

Note: *p<0.1; **p<0.05; ***p<0.01.

Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton, merged at (*year, vessel, region*) resolution with fallback to year-level national mean when the regional cell is missing.

Allocated harvest is species-specific: $H_{vy,s}^{alloc} = \omega_{vs}^{hist} \bar{Q}_{sy}^{r(v)}$ following Kasperski (2015) Eq. 17. The artisanal regression interacts $H_{vy,anchoveta}^{alloc}$ with TIPO_EMB; TIPO_EMB categories with $N < 50$ vessel-years (BR, BRV, L) are retained in the fit for completeness but excluded from the aggregate factor_trips projections.

Year fixed effects absorb aggregate shocks (estallido social 2019, COVID 2020–2022).

LR test NB vs Poisson: Industrial $\chi^2 = 397.6$; Artisanal $\chi^2 = 37046.9$ (both $p < 0.001$).

toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine–anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The anchoveta-price coefficient is negative and significant; this counterintuitive sign likely reflects simultaneity: years of low anchoveta availability generate both higher prices (through the inverse-demand channel) and fewer trips (through reduced catch opportunities), producing a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an empirically grounded mapping for the simulation, in which prices are determined endogenously by the inverse-demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but it is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns across fleets. For the artisanal fleet, adverse weather days carry a negative and statistically distinguishable coefficient, indicating that, controlling for year-specific aggregate conditions, vessels operating in grids with more days of wind speeds above 8 m/s undertake fewer fishing trips. For the industrial fleet, adverse weather is not significant, consistent with larger vessels being better equipped to operate under rough conditions. Biological closure days are strongly negative for the industrial fleet, representing the largest effect in the model, consistent with the industrial fleet’s dependence on access during open seasons. For the artisanal fleet, the positive coefficient on closure days does not admit a causal interpretation: under the year-fixed-effects specification, β_{closure} is identified solely from the cross-zone differential between regions V-VIII (151 days) and IX-XIV (182 days) of the proxy variable described in Online Appendix A and discussed at length in the regulatory-history caveat of Section 5. The positive sign therefore reflects locational heterogeneity in artisanal fishing intensity between the two regulatory zones (the southern zone IX-XIV contains both more nominal closure days and more concentrated artisanal effort per vessel, and the cross-sectional comparison cannot disentangle the two), not the causal effect of closures on trips. Constructing a year-by-year closure variable

with within-zone temporal variation that would identify the causal effect requires reconciling the nominal regulatory ceilings of the D.Ex. 115/1998 and 530/2016 chain with the actual realised closure days reported in the SUBPESCA Indicadores Biológicos portal from 2019 onwards, a non-trivial data-engineering task discussed in Section 5 and reserved for follow-up work.

Vessel-type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

4.3 Climate change projections

The projection methodology is summarized in Section 3.4. Under each scenario and time window, we translate CMIP6-derived environmental deltas into (i) changes in bad-weather days that enter the trip equation (direct channel) and (ii) posterior draws of the stock-specific effective growth rate $r_{i,t}^*$, obtained by evaluating the identified climate shifter of Eq. (2) at the projected SST and log-CHL anomalies (indirect channel).

4.3.1 Projected environmental changes

Table 4 summarises the projected environmental changes as cross-model medians with inter-model IQR. SST warms by +0.7 to +2.4°C across scenarios. The chlorophyll-a signal is small in magnitude relative to cross-model spread, reflecting documented CMIP6 disagreement on primary-productivity responses. Wind speed increases modestly (+0.1 to +0.4~m/s cross-model median), but translates into +10 to +23.5 additional bad-weather days per year because the historical daily wind distribution at vessel COGs sits sufficiently close to the 8~m/s operability threshold that a small upward shift in the mean displaces a non-negligible mass of the upper tail past it.

Table 4: Projected environmental changes for Centro-Sur Chile (CMIP6 six-model ensemble, delta method; cross-model median with inter-model IQR in brackets).

SSP	Window	Δ SST (°C)	Δ CHL (%)	Δ wind speed (m/s)
SSP2-4.5	2081–2100	+1.44 [+0.96, +1.91]	+0.0 [−3.6, +1.4]	+0.19 [+0.10, +0.24]
SSP2-4.5	2041–2060	+0.74 [+0.52, +0.92]	+2.2 [+0.0, +2.8]	+0.14 [+0.13, +0.15]
SSP5-8.5	2081–2100	+2.39 [+1.93, +3.29]	−1.3 [−7.4, +3.0]	+0.42 [+0.36, +0.45]
SSP5-8.5	2041–2060	+0.92 [+0.61, +1.23]	+0.7 [−2.3, +3.4]	+0.21 [+0.14, +0.32]

4.3.2 Comparative statics on intrinsic stock productivity

Table 5 reports the comparative-statics object $\Delta r_i^*/r_i^0$, obtained by evaluating the identified climate shifter at the CMIP6 ensemble deltas and integrating over the posterior of $(\rho_i^{SST}, \rho_i^{CHL})$. We report two distinct uncertainty axes separately: the cross-model median and IQR of the model-level posterior medians (climate uncertainty), and the median across models of the within-model 90% posterior credible interval (structural uncertainty); a separate variance decomposition appears in Online Appendix F. Jurel is reported as non-identified (“n.i.”) consistent with the identification result in Section 4.1.

Table 5: Comparative statics on intrinsic stock productivity under the CMIP6 six-model ensemble.

Stock	Scenario	n_m	ΔSST (°C)	$\Delta \log CHL$	$\Delta r_i^*/r_i^0$ (cross median)	Cross-model IQR	Within posterior 90% CI	$\Pr(\Delta < 0)$
Anchoveta CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-50.7%	[-52.7%, -47.8%]	[-68.0%, -27.0%]	1.00
Anchoveta CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-71.0%	[-76.3%, -60.4%]	[-91.1%, -31.0%]	1.00
Anchoveta CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-62.4%	[-68.0%, -55.9%]	[-80.9%, -31.2%]	1.00
Anchoveta CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-89.6%	[-91.5%, -87.4%]	[-98.3%, -54.2%]	0.99
Sardine CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-78.6%	[-93.2%, -72.5%]	[-85.7%, -67.3%]	1.00
Sardine CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-97.2%	[-98.9%, -89.8%]	[-98.9%, -92.6%]	1.00
Sardine CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-91.1%	[-96.7%, -78.5%]	[-95.1%, -83.2%]	1.00
Sardine CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-99.9%	[-100.0%, -99.4%]	[-100.0%, -99.3%]	1.00
Jack mackerel CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	n.i.	n.i.	n.i.	n.i.

Note:

Entries summarise the posterior of the proportional change in intrinsic productivity, $\Delta r / r_0 = \exp(\rho_{SST} * \Delta SST + \rho_{CHL} * \Delta \log CHL) - 1$, evaluated at scenario-window-specific CMIP6 anomalies for each of n_m models in the ensemble and integrated over the posterior of (ρ_{SST}, ρ_{CHL}) from the full state-space specification. The cross-model IQR is the 25th-to-75th percentile range across CMIP6 models of the within-model posterior medians and captures inter-model climate uncertainty. The within posterior 90% CI is the cross-model median of the within-model 5th-to-95th-percentile posterior interval and captures structural uncertainty about (ρ_{SST}, ρ_{CHL}) holding the climate forcing fixed. Jurel (jack mackerel) CS is reported as non-identified (“n.i.”) because the posterior of its shifters coincides with the weakly informative prior; see the identification discussion in Results. $n_m = 5$ instead of 6 in the SSP2-4.5 panels because CESM2 does not publish chl-ssp245 in the NCAR CMIP6 catalogue. Under SSP5-8.5 end-of-century the implied SST anomaly lies approximately nine in-sample standard deviations outside the realised 2000-2024 envelope, so it is best read as a structural-form stress test under the maintained log-linear shifter. Raw numerical values are archived in tables/growth_comparative_statics_raw.csv; model-level disaggregation is in tables/growth_comparative_statics_by_model.csv. Appendix F reports the variance decomposition between cross-model and within-model components.

The magnitudes are economically first-order for a Centro-Sur fishery whose combined TAC is in the low hundred-thousand-tonne range. Under the moderate SSP2-4.5 mid-century scenario (cross-model median +0.74°C; $\Delta \log CHL$ near zero in the cross-model median, with non-trivial inter-model spread), the cross-model median of anchoveta’s intrinsic productivity declines by 51% and sardina común’s by 79%, with posterior probability of negative impact equal to one in both cases. Under the high-emissions SSP5-8.5 end-of-century scenario (cross-model median +2.39°C, $\Delta \log CHL$ cross-model median −0.013 with 5th-percentile of −0.30 across models) the cross-model medians deepen to −90% and −99.9% respectively, and the entire 90% credible interval of sardine’s response lies below zero in every model of the ensemble; the anchoveta cross-model probability of

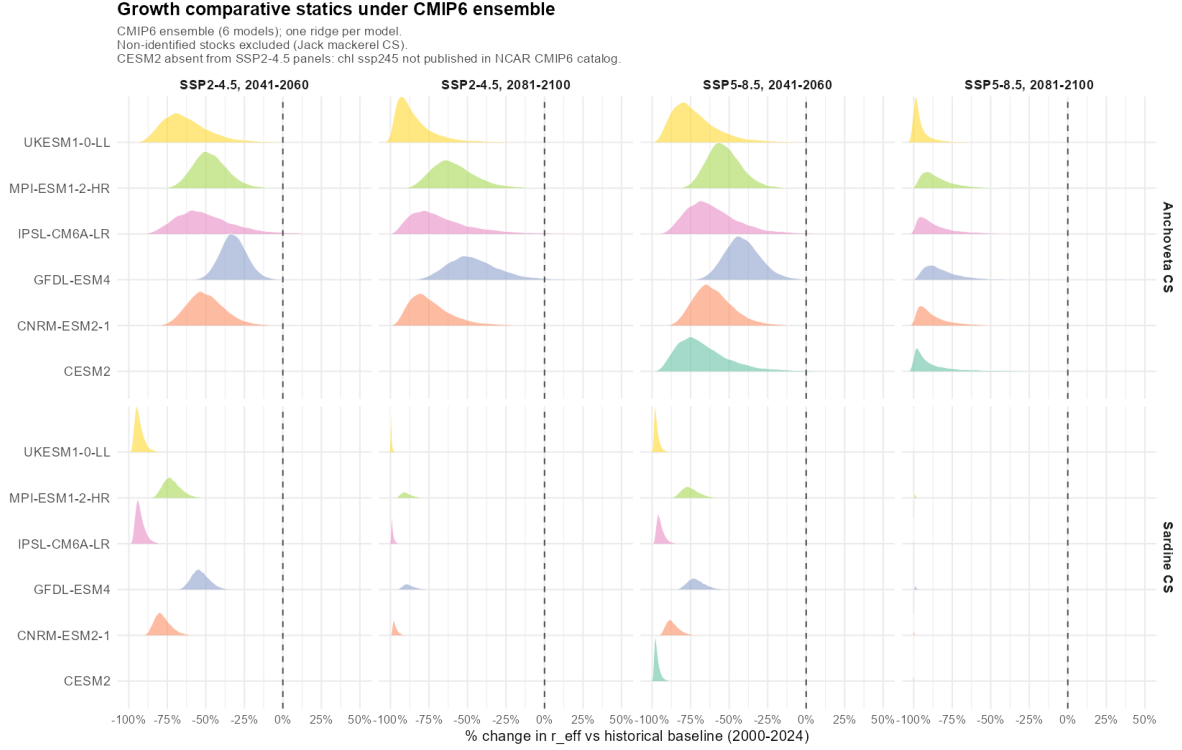


Figure 4: Posterior distribution of the proportional change in intrinsic stock productivity under CMIP6 scenarios, six-model ensemble. Ridges are kernel-smoothed marginal posteriors of the proportional change in intrinsic productivity derived from draws of the state-space full specification, with one ridge per CMIP6 model in each scenario–window pair; vertical line at zero. CESM2 is absent from the SSP2-4.5 panels because chl-ssp245 is not published in the NCAR CMIP6 catalogue. Jack mackerel is omitted because its climate shifter is not identified in the 2000–2024 sample.

decline is 0.99 rather than 1 because under a small fraction of (model, draw) pairs a sufficiently negative $\Delta \log \text{CHL}$ combined with $\rho_{\text{anch}}^{\text{CHL}} < 0$ offsets the SST-driven collapse. The identified signs confirm the qualitative reading of the shifters in Section 4.1: both coastal stocks respond negatively to SST, but sardina común carries roughly twice the SST semi-elasticity of anchoveta, and its CHL semi-elasticity is of the opposite sign, so that the compound response—negative thermal shock combined with negative primary-productivity shock—is sharper for sardina than for anchoveta.

The comparative-statics exercise is a long-run object rather than a year-by-year forecast. It describes the steady-state intrinsic productivity of each stock under a counterfactual climate, *holding fixed* the law of motion and the fishery’s regulatory regime. The shifters $(\rho_i^{\text{SST}}, \rho_i^{\text{CHL}})$ enter the law of motion as coefficients on exogenous climate forcings, which makes them invariant to the joint distribution of (SST, B, C) and therefore evaluable at climate regimes outside the estimation

window—in particular, at an SST anomaly of $+2.3^{\circ}\text{C}$ that lies roughly nine historical standard deviations beyond the 2000–2024 mean. A purely correlational specification would not license this extrapolation.

4.3.3 Implications for fleet-level effort

Table 6 translates the comparative-statics object on intrinsic stock productivity in Table 5 into a comparative-statics object on fleet-level annual trips, by propagating each posterior draw through the stock-dynamics steady-state equation under historical average fishing pressure and through the negative binomial trip equation of Section 4.2. Vessel-level responses are aggregated to the fleet level as medians over posterior draws and over vessels within fleet, weighted by each vessel’s realized catch composition over the 2012–2024 estimation window.

The posterior probability of portfolio loss—defined as a reduction of more than fifty percent in the composition-weighted biomass factor f_v^H relative to the 2000–2024 historical baseline—exceeds 0.95 for the artisanal fleet under every CMIP6 scenario considered and reaches 0.99 under SSP5-8.5 end-of-century. For the industrial fleet, the loss probability is a stable 0.12 across all four scenarios, driven entirely by its five-percent exposure to sardina común; the remaining 95% of its historical portfolio is allocated to jack mackerel, whose climate shifter is treated as non-identified in Section 4.1. The asymmetry in loss probability between the two fleets is first-order and stable across climate pathways, approximately eight-fold across all four scenarios.

Decomposing the fleet-level trip response isolates the channels through which this asymmetry operates. The *marginal* posterior median of $\%\Delta$ trips ranges from -13.2% (SSP2-4.5 mid) to -15.0% (SSP5-8.5 end) for the artisanal fleet and from -0.8% to -1.0% for the industrial fleet—a ratio of roughly fifteen to one at end-of-century. The *conditional* posterior median, restricted to draws with $f_v^H \geq 0.5$, isolates the climate-elasticity component from the portfolio-collapse component and yields -3.8% to -4.6% for artisanal and -0.7% to -1.0% for industrial, a conditional ratio of roughly four to one. Two channels generate the asymmetry. The *indirect biomass channel* transmits the climate signal asymmetrically through portfolio composition: the artisanal fleet is concentrated in the two coastal-upwelling stocks with identified climate shifters, while 95% of the industrial fleet’s portfolio sits in jurel, whose Centro-Sur shifter is non-identified and held at the

Table 6: Long-run implications of CMIP6 climate scenarios for fleet-level fishing effort, under a Schaefer steady-state thought experiment with fishing pressure fixed at the 2000–2024 historical average. Entries are cross-model medians with cross-model IQRs in brackets, aggregated within each CMIP6 model over posterior draws and vessels within fleet.

Fleet	Scenario	$\Pr(f_v^H < 0.5)$	% Δ trips (marginal)	% Δ trips (conditional)
Artisanal	SSP2-4.5, 2041-2060	0.95 [0.93, 0.96]	-13.2% [-13.3%, -12.2%]	-4.6% [-4.8%, -4.0%]
Artisanal	SSP2-4.5, 2081-2100	0.98 [0.97, 0.99]	-14.1% [-14.3%, -13.7%]	-3.9% [-4.5%, -3.4%]
Artisanal	SSP5-8.5, 2041-2060	0.98 [0.96, 0.98]	-13.6% [-13.8%, -13.4%]	-4.2% [-4.5%, -4.1%]
Artisanal	SSP5-8.5, 2081-2100	0.99 [0.99, 0.99]	-15.0% [-15.2%, -14.9%]	-3.8% [-5.5%, -2.6%]
Industrial	SSP2-4.5, 2041-2060	0.12 [0.11, 0.12]	-1.0% [-1.0%, -0.8%]	-0.7% [-0.7%, -0.7%]
Industrial	SSP2-4.5, 2081-2100	0.12 [0.12, 0.12]	-0.9% [-1.0%, -0.8%]	-0.7% [-0.8%, -0.7%]
Industrial	SSP5-8.5, 2041-2060	0.12 [0.12, 0.12]	-0.8% [-0.9%, -0.8%]	-0.7% [-0.9%, -0.7%]
Industrial	SSP5-8.5, 2081-2100	0.12 [0.12, 0.12]	-1.0% [-1.1%, -1.0%]	-1.0% [-1.0%, -0.9%]

Note:

Entries summarise the posterior of the proportional change in fleet-level annual trips, computed for each of the six CMIP6 models in the ensemble and then aggregated across models. Within each model, the posterior is integrated over draws of the full state-space specification and vessels within fleet (composition-weighted exposure). Cross-model medians and 25th-to-75th-percentile inter-quartile ranges across the n_m models are reported in brackets; the median across models of the within-model 5th-to-95th-percentile posterior interval (within posterior CI) is archived in tables/trip_comparative_statics_raw.csv. f_H is the composition-weighted biomass factor relative to the historical baseline: $f_H = \text{sum over species of } \omega_{vs} * B_{star_s} / B_{hist_s}$, where $B_{star_s} = K_s * (1 - F_{hist_s} / r_{eff_s})$ is the Schaefer steady-state biomass and ω_{vs} is the vessel’s realized catch share in species s . $\Pr(f_H < 0.5)$ reports the posterior probability that the vessel’s composition-weighted biomass factor falls below one-half. The marginal median integrates over all posterior draws within each model; the conditional median restricts to draws with $f_H \geq 0.5$, isolating the pure climate-elasticity component from the steady-state collapse component of the trip response. Jack mackerel is treated as non-identified ($\text{factor}_B = 1$) consistent with the identification discussion in the Results section. Raw numerical values, by-model disaggregation, and by-stock extinction probabilities are archived in tables/trip_comparative_statics_raw.csv, tables/trip_comparative_statics_by_model.csv, and tables/trip_comparative_statics_extinct.csv. The narrow cross-IQR observed in many cells reflects a floor effect: once Schaefer steady-state collapse saturates the portfolio-weighted biomass factor across models, the trip response converges to $\exp(-\beta * H_{alloc} * 1)$, independent of climate magnitude. Appendix G reports the formal two-way variance decomposition of the fleet-level trip response, which confirms that the within-model component (posterior plus vessel heterogeneity) dominates the total variance under saturated regimes; Appendix F reports the analogous decomposition for the underlying productivity response.

climatological mean. The *direct weather channel* transmits the signal through differential weather sensitivity: $\hat{\beta}_{\text{weather}}^{\text{ART}} = -0.0007$ (marginally significant) vs $\hat{\beta}_{\text{weather}}^{\text{IND}} = -0.0003$ (indistinguishable from zero), translating the projected +10 to +23.5 days of additional bad weather into a -0.7% to -1.9% contribution to the artisanal response and near-zero for industrial. The limited cross-sector transferability of the prevailing sector-level allocation locks this dual exposure into differential losses of harvest capacity, consistent with the portfolio mechanism documented by Kasperski & Holland (2013) and the access constraints of Oken et al. (2021).

The between-fleet asymmetry decomposes further into within-artisanal heterogeneity by vessel

type (Table 7). The dominant *lancha motor* (LM) segment (77% of the panel) contracts by -19.9% under SSP5-8.5 end-of-century, driven by its identified positive sardine elasticity and the projected -99.9% steady-state collapse of the sardine stock. The smaller *bote motor* (BM) segment (8%) contracts by only -3.2% at the marginal median, because its identified positive anchoveta elasticity ($\hat{\beta}_h^{\text{anch,BM}} = +0.016$, $p < 0.001$) partially offsets the sardine contraction in the high-mass posterior region. The intermediate UNK segment (14%) contracts by -7.6% . Portfolio-loss probability exceeds 0.98 in all three categories, so the differential within artisanal is concentrated in the magnitude rather than the incidence of portfolio collapse.

Table 7: Within-artisanal-fleet decomposition of the long-run trip response by vessel type, under SSP5-8.5 end-of-century. Entries are cross-model medians with cross-model IQRs in brackets. The marginal column integrates over all posterior draws; the conditional column restricts to draws in which the composition-weighted biomass factor is at least one-half, isolating the climate-elasticity component from the steady-state collapse component.

	Segment	Pr(portfolio loss)	% Δ trips (marginal)	Cross-model IQR	% Δ trips (conditional)
LM	Lancha motor (LM)	0.99	-19.9%	[-20.0%, -19.7%]	-16.1%
UNK	Unidentified (UNK)	0.98	-7.6%	[-8.1%, -7.3%]	-1.6%
BM	Bote motor (BM)	1.00	-3.2%	[-3.3%, -3.0%]	+16.3%

Note:

Categories with fewer than 50 vessel-year observations (BR, BRV, L) collectively account for less than 0.1 percent of artisanal effort and quota and are excluded from this table. Vessel type BM is the reference category in the negative binomial interaction; LM and UNK enter as deviations. Raw cross-model values and other scenario-window combinations are archived in `trip_comparative_statics_by_tipo_emb_raw.csv`.

Three caveats qualify this thought experiment. First, the Schaefer steady-state comparative statics holds fishing pressure fixed at the 2000–2024 historical average; any regulatory response that lowered F in the face of falling productivity would attenuate both the loss probability and the marginal trip response reported here. Quantifying that regulatory margin requires a forward simulation of the biomass process under endogenous harvest rules—most prominently the LTP cap on industrial quota transferability and the annual SUBPESCA revision of TACs in response to in-year biomass signals—which we leave to future work and which would reuse the posterior reported here. Second, the 2000–2024 observation window was not itself in Schaefer equilibrium under F_{hist} ; the pipeline’s internal consistency check, obtained by evaluating the comparative statics at $\Delta SST = 0$ and $\Delta \log CHL = 0$, yields a median f_v^H of approximately 1.015 rather than 1.000, reflecting a mild historical depletion from equilibrium. The scenario results in Table 6 should therefore be read as responses *relative to the baseline steady-state* rather than relative to the 2000–2024 observed

trajectory. Third, the jack mackerel shifter is treated as non-identified. Two interpretive points should precede the empirical evidence. First, the result is a null on *local identification*, not on biological climate sensitivity: the available 25-year window provides only 16 informative annual observations of Centro-Sur spawning biomass against a historical SST envelope of $\hat{\sigma}_{SST} \approx 0.26^\circ\text{C}$, so any climate response of moderate magnitude operating at scales broader than the Centro-Sur biological domain would lie below the threshold of statistical detection in this sample regardless of its true magnitude. Second, complementary empirical evidence on jack mackerel documents that climate forcing does operate on the species at scales beyond local coastal anomalies: Peña-Torres et al. (2017) report ENSO-driven shifts in jack mackerel location choices in the Chilean fishery, and Arcos et al. (2001) describe SPRFMO-scale stock dynamics consistent with broad-basin forcing, so the absence of an identifiable Centro-Sur shifter should not be read as evidence of climate insensitivity. With these interpretive caveats noted, four converging tests reported in Online Appendix E converge on the same null and rule out the most natural mechanical explanations for it—insufficient spatial extent, insufficient sample size, poor source aggregation, and a poor choice of spatial scale for the climate shifter. Re-fitting the state-space on three nested spatial domains for the environmental aggregation (Centro-Sur EEZ, offshore-extended, regional Southeast Pacific) yields posterior-to-prior standard deviation ratios of 0.998–1.014 for both $\rho_{\text{jurel}}^{SST}$ and $\rho_{\text{jurel}}^{CHL}$, indistinguishable from unity in every domain (Online Appendix E.2). A dual-source extension that augments the likelihood with the Northern Chilean acoustic series under a common climate elasticity—an assumption motivated by the 0.88 log-scale correlation between the Centro-Sur and Northern series across overlapping years—reduces the ratios only to 0.94–0.99, well above the threshold for material identification (Online Appendix E.3). Replacing the local coastal shifters with the basin-scale ENSO Niño 3.4 index, a test of whether the relevant climatic forcing for jack mackerel operates at the basin scale rather than the coastal scale, returns a posterior-to-prior standard deviation ratio of approximately 0.98 at lag one, indistinguishable from unity (Online Appendix E.4). A separate exploratory attempt to incorporate the SPRFMO range-wide assessment (OROP-PS, encompassing Chile, Peru, Ecuador and the adjacent high-seas area) fails on coherence grounds, with multimodal posterior support and unsatisfactory chain mixing, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter. The convergence of these four tests indicates a genuinely flat shifter *within the Centro-Sur biological*

domain, consistent with the most natural reading of the null in light of Peña-Torres et al. (2017) and Arcos et al. (2001): that the relevant climatic drivers for jack mackerel operate at spatial scales beyond those captured by local Chilean SST and chlorophyll anomalies. The industrial fleet’s apparent climate protection in Table 6 therefore rests on a null about local identification that cannot be sharpened within the Centro-Sur identification strategy, even with the dual-source extension; the appropriate setting in which to discriminate between local coastal decoupling and broader-scale climate forcing is a range-wide SPRFMO analysis with explicit cross-stock spillover, which we leave to future work.

5 Discussion

Our results reveal a substantial asymmetry in how climate change affects different segments of Chile’s small pelagic fishery. The asymmetry is governed by the differential exposure of each fleet’s species portfolio to the identified climate parameters under the limited cross-sector transferability of the prevailing sector-level allocation. The two coastal upwelling stocks targeted by the artisanal fleet—anchoveta and sardina común—both respond negatively to warming, with sardina común carrying roughly twice the SST semi-elasticity of anchoveta and an opposite-signed CHL response. The industrial fleet, by contrast, is diversified across these two coastal stocks and the transboundary jack mackerel stock, whose local productivity is *not* identified in the 2000–2024 sample. The artisanal fleet, concentrated in the sardina–anchoveta pair, is therefore the segment whose long-run harvest capacity is most exposed to the CMIP6 climate signal under the prevailing sector-level allocation. This finding aligns with the broader literature on heterogeneous climate impacts within fleets (Free et al., 2019; Sumaila et al., 2011) and underscores that aggregate projections can mask first-order distributional consequences inside the same fishery. The asymmetry further decomposes within the artisanal fleet by vessel type (Table 7): the dominant LM segment contracts by approximately twenty percent (driven by sardine collapse), while the smaller BM segment contracts by only three to four percent (its positive identified anchoveta elasticity partially offsetting the sardine signal), so the between-fleet asymmetry that anchors the headline result is itself the aggregate of a between-fleet and a within-artisanal mechanism that operate through the same portfolio channel.

The non-identification of $(\rho_{\text{jur}}^{SST}, \rho_{\text{jur}}^{CHL})$ at the Centro-Sur scale is itself a substantive result rather than a nuisance, but its policy reading depends on a careful distinction between *non-identification* and *no climate effect*. The two are easily conflated by specifications that force a point estimate onto every regressor regardless of how informative the data are about it. Online Appendix E reports five converging tests of this distinction, all consistent with the absence of an identifiable shifter for jack mackerel within the 2000–2024 Centro-Sur biomass record: (i) varying the spatial aggregation of the coastal shifters across three nested domains $\mathcal{D}_1 \subset \mathcal{D}_2 \subset \mathcal{D}_3$ leaves the posterior-to-prior ratios in 0.998–1.014; (ii) augmenting the biomass record with a Northern Chilean acoustic series under a common elasticity moves the ratio only to 0.94–0.99; (iii) replacing the coastal shifters with the basin-scale ENSO Niño 3.4 index returns a ratio of 0.98 at lag one and 1.01 at lag two; (iv) a joint specification with all three shifters active for jack mackerel simultaneously returns ratios of 1.03 on local SST, 1.00 on local log chlorophyll, and 0.98 on basin-scale ENSO; and (v) an attempt to incorporate the SPRFMO regional assessment series fails on coherence grounds. Existing empirical evidence on jack mackerel documents that climate does affect the species at scales broader than local coastal [anomalies—@Arcos2001-jq](#) on intrusion of the 15 °C isotherm during the 1997–98 El Niño, Peña-Torres et al. (2017) on ENSO-driven shifts in location choices in the Chilean fishery—but each of these documents climatic effects on *margins* (location choice, range distribution, spawning timing, phenological response) that are not necessarily captured in the elasticity of structural productivity at an annual aggregate. The convergence of five independent tests on this null at the Centro-Sur scale points to a structural rather than aggregation-driven limitation of the available record: closing the identification gap will require either a longer biomass record or a basin-scale assessment with explicit cross-stock spillover, not a different choice of spatial aggregation, biomass source, basin-scale shifter, or joint-specification convention within the present record. We accordingly hold r_{jurel}^* fixed in the projections of Tables 5 and 6, and report the resulting fleet-level elasticities under that constraint. A prior-propagation sensitivity reported in Online Appendix E.4 computes the 90% predictive interval of the jack mackerel productivity factor by combining the posterior of ρ_{jur}^{ENSO} with the CMIP6 ensemble of basin-scale temperature deltas; the interval spans approximately three orders of magnitude under SSP5-8.5 end-of-century ([0.05, 19.5] on the productivity factor), confirming that the prior-propagated projection is not informative for policy and that the fixed- r^* convention is the defensible reporting choice.

The two channels through which climate change reaches fleet-level effort operate on very different scales and on different segments of the fleet. The direct weather channel is small in mean wind anomaly—the cross-model median CMIP6 increase in near-surface wind speed is below +0.5 m/s even under SSP5-8.5 end-of-century—but the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that additional bad-weather days exceed twenty per year on average under SSP5-8.5 end-of-century. This translates into a -0.7% to -1.9% marginal contribution to the artisanal trip response and a contribution indistinguishable from zero for the industrial fleet, reflecting the differential weather sensitivity captured by $\hat{\beta}_{\text{weather}}$. The indirect biomass channel dominates the response in absolute terms for both fleets but operates entirely through portfolio composition, so the asymmetry it generates is intrinsic to the fixed sector-level allocation of stocks across segments. Climate adaptation policy in the Centro-Sur therefore needs to address both margins. The indirect channel calls for reform of the institutional architecture that mediates portfolio exposure—quota allocation rules, cross-fleet transferability, and the design of TACs under shifting species productivity. The direct channel calls for operational measures targeted at the artisanal segment specifically, such as expanded port infrastructure, vessel reinforcement programs, or weather-indexed insurance, which are first-order for that fleet but redundant for the industrial purse-seine fleet. This conclusion is based on the CMIP6 monthly wind ensemble; downscaled regional climate models that resolve extreme wind events could only raise the weight of the direct channel further.

The species-level productivity projections deserve careful interpretation. The log-linear form of the climate shifter in Eq. (2) is identified from interannual variation over 2000–2024, during which the realised range of SST anomalies was narrow ($\hat{\sigma}_{SST} \approx 0.26^\circ\text{C}$). CMIP6 anomalies under SSP5-8.5 end-of-century reach roughly $+2.3^\circ\text{C}$, nearly an order of magnitude outside the estimation window, so the corresponding posterior of $\Delta r_i^*/r_i^0$ leans heavily on the log-linear extrapolation embedded in the shifter function. We regard the projected magnitudes as indicative of the direction and relative scale of impacts across stocks rather than as precise point forecasts. The log-linear shifter is a maintained simplification: thermal-tolerance arguments in the climate-fisheries literature (Cheung et al., 2010; Free et al., 2019) imply that for stocks operating below their thermal optimum, a log-linear specification, if anything, *under-states* the productivity response under high warming.

The projected magnitudes for SSP5-8.5 end-of-century should therefore be read as conservative within the class of monotone shifter functions consistent with the identified posterior signs. A non-parametric extension of the shifter, or pooled identification across multiple Humboldt-Current datasets, is the natural direction in which to relax this assumption.

Seven caveats qualify the present analysis: holding ex-vessel prices and management rules constant; substantial cross-model spread in chlorophyll-a; identification on a 25-year series limiting non-linearity detection; the absence of risk preferences in the trip equation; the zonal coding of biological closures; the purse-seine focus of the trip-equation panel; and the missing wind components for CESM2 in the direct weather channel. Online Appendix I states each in full and qualifies its implication for the interpretation developed above.

The finding that climate change creates winners and losers within the same fishery has direct implications for quota-allocation policy. A natural first reading would attribute the asymmetry to rigid single-species permits in the artisanal sector, but both sectors operate *de jure* multi-species permit portfolios: SUBPESCA's 2012 registry records simultaneous authorisations across all three pelagic species for 76% of industrial vessels ([Subsecretaría de Pesca y Acuicultura \(SUBPESCA\), 2026](#)); the IFOP logbook in turn reveals that 90% of artisanal vessels operate a multi-species portfolio in any given year. The asymmetric climate sensitivity we identify is therefore not driven by permit count but by four dimensions of the institutional architecture. First, the statutory partition fixes inter-sectoral shares at the legislative level (Section~??, Table~1). Second, the partition admits a single administratively rationed re-allocation channel under Article₅₅T: industrial titulares cede 85–99% of their assigned anchoveta and sardina com'un TACs to artisanal cessionaries each year (25–77~kt per species-year), while jurel cessions are an order of magnitude smaller and reverse direction across years (Online Appendix H.3). The channel is non-market in the conventional sense but operational rather than dormant. Article~6 of Ley~21.752 imposes a 1.3~UTM/ton patent on industrial-to-artisanal cessions, re-pricing the pre-existing channel rather than introducing a new wedge. Third, within-sector transferability is asymmetric: industrial coefficients trade under LTP since 2013, while artisanal RAE assignments remain locked to the regional registry. Fourth, the revealed portfolio composition differs in its climate correlation structure: the artisanal logbook shows 80–86% of activity concentrated in the anchoveta–sardina pair (both with identified negative

climate elasticities), while 67% of industrial activity sits in the same pair with the remainder in jurel and caballa (non-identified at the Centro-Sur scale). The asymmetry the paper estimates therefore quantifies the cost of a statutory partition combined with a climate-correlated artisanal portfolio under projected SSP5-8.5 stress. Online Appendix H reports the descriptive evidence and Online Appendix H.2 computes a policy counterfactual.

5.1 Implications of the 2025 fractioning reform

The asymmetry estimated in this paper is identified on the Article₄₇LGPA fractioning regime that prevailed during the estimation window (2018–2024). Ley_{Nº}21.752, in force from 2026 until 2040, modifies that regime in two relevant directions for the Central-South small pelagic fishery ([Congreso Nacional de Chile, 2025](#)). First, the artisanal share of the anchoveta and sardina com'un TACs rises from 78% to 90% across the V–Los–Lagos macrozone; the same artisanal sector therefore now holds a deeper claim on stocks whose long-run productivity is projected to collapse under SSP5-8.5 end-of-century. Second, the artisanal share of the jurel TAC rises from 10% to 30% in V–Los~R'ios and from 10% to 15% in Los~Lagos—a quota-weighted aggregate of approximately 28% across the macrozone—giving the artisanal sector partial access to the species whose local productivity is non-identified at the Centro-Sur scale and whose climate response is held fixed in our projections. Table~8 reports the static plug-in counterfactual at SSP5-8.5 end-of-century, holding the structural posterior and the TAC level at 2024 values and varying only the statutory artisanal share.

Table 8: Static counterfactual: artisanal absolute landing under the Article 47 LGPA and Ley 21.752 fractioning regimes at SSP5-8.5 end-of-century, Centro-Sur (V–Los Lagos), 2024 Cuotas Globales.

Stock	TAC 2024 (kt)	Artisanal share (α_s)		Artisanal landing L_{art} (kt)	
		Art. 47	Ley 21.752	Art. 47	Ley 21.752
Anchoveta (V–Los Lagos)	211.3	0.78	0.90	17.1	19.8
Sardina común (V–Los Lagos)	290.6	0.78	0.90	0.2	0.3
Jurel (V–Los Ríos)	578.6	0.10	0.30	57.9	173.6
Jurel (Los Ríos–Los Lagos)	80.6	0.10	0.15	8.1	12.1
Total Centro-Sur				83.3	205.7

Notes. Static plug-in evaluation of $L_{\text{art}}^{(R)} = \sum_s \alpha_s^{(R)} \cdot (1 + \widehat{\Delta r_s^* / r_s^0}) \cdot Q_s^0$, with $\widehat{\Delta r_s^* / r_s^0}$ set to the cross-model median at SSP5-8.5 end-of-century from Table 5 (−89.6% anchoveta, −99.9% sardina común, 0% jurel by the non-identification convention of Section 3.3.1). Q_s^0 is the 2024 Cuota Global de Captura. Artisanal fractions follow Article 47 LGPA (pre-2025) and Article 1 of Ley 21.752. The Los Ríos–Los Lagos jurel quota is reported as a single block by SERNAPESCA; the row applies the Los Lagos artisanal share (15%) as a conservative lower bound on the true sub-block share, which would rise toward 30% to the extent that Los Ríos contributes to the aggregate quota. The posterior-propagated version of this table, with cross-model IQR and within-posterior 90% CI, is reported in Online Appendix H.2.

Three readings of Table~8 structure the policy interpretation. First, the redistribution *raises* the absolute artisanal landing under projected SSP5-8.5 end-of-century stress by approximately 122~kt at the Centro-Sur aggregate, almost entirely through the 20~percentage-point increase in the artisanal jurel share in V–Los-R'ios rather than through the small pelagics. Because the projected productivity of the anchoveta–sardina pair collapses under the same scenario, the 12~percentage-point increase in the artisanal share of those stocks operates on a vanishing base and is therefore mechanically small in absolute terms. Second, the policy logic of the redistribution—which was designed to protect the artisanal sector from over-allocation to the industrial fleet—in fact *deepens* the artisanal exposure to small-pelagic biomass collapse under climate stress: the same artisanal sector now holds 90% rather than 78% of a TAC that the structural posterior projects to fall to floor levels. The net welfare effect at SSP5-8.5 end-of-century therefore depends on whether the jurel access expansion outweighs the deepened small-pelagic exposure, a question that the

present static counterfactual cannot answer without an inverse-demand system and a forward-looking allocation rule. Third, the static counterfactual reported in Table~8 should be read as a de jure upper bound on the operational re-allocation produced by Ley~21.752, not as the realised effect on sectoral landings. Under the pre-2025 regime, the industrial sector ceded 85–99% of its assigned anchoveta and sardina com'un TACs to artisanal cessionaries via Article~55T (Online Appendix H, Online Appendix H.3), so the artisanal sector's effective access to the small-pelagic biomass already approached the de jure ceiling of unity. The 12~percentage-point rise in the artisanal de jure share therefore translates into a much smaller change in realised artisanal landings; the operationally meaningful change in those two stocks is the contraction of the cession volume that industrial titulares had been intermediating for fee, together with the 1.3~UTM/ton patent that Article~6 now levies on the residual cession flow. The reform is therefore primarily *redistributive*—it transfers the rent extracted on the Article~55T channel from industrial titulares to artisanal armadores—rather than *allocative* in the small pelagics. For jurel the calculus differs: historical cessions are an order of magnitude smaller relative to the TAC, and the 20 and 5~percentage-point increases in artisanal de jure shares under items~4 and~5 of Article~1 correspond to an operationally meaningful expansion of artisanal access. A full welfare evaluation of the Ley~21.752 regime under climate stress—one that endogenises the realised cession volume as a function of the patent rate and projected stock productivity—is left as the natural extension developed in the Conclusions; the present static plug-in counterfactual quantifies the de jure boundary case from which that endogenous response should be measured.

6 Conclusions

This paper identifies the structural climate elasticities of stock productivity in the Chilean small pelagic fishery and combines them with a vessel-level effort equation to compute long-run comparative statics on fleet-level fishing effort under projected climate paths. We contribute to the literature in three ways. First, we couple a Bayesian state-space model of multi-species stock dynamics—calibrated on official IFOP and SPRFMO assessments—with negative binomial trip equations estimated separately for the artisanal and industrial fleets, and we show that the direction of the distributional asymmetry across fleets is governed by the interaction between portfolio

composition and the limited cross-sector transferability of the prevailing sector-level allocation. The artisanal segment bears the sharper long-run loss because its historical landings are concentrated in the two stocks whose climate sensitivities are identified and economically large. This finding extends the portfolio-and-access mechanism of Kasperski & Holland (2013), Cline et al. (2017), and Oken et al. (2021) to a climate-shock setting under a fixed industrial-artisanal allocation. Second, we identify a vector of climate semi-elasticities ($\rho_i^{SST}, \rho_i^{CHL}$)—the structural parameters that translate sea-surface temperature and primary-productivity anomalies into shifts in long-run stock productivity—recovered sharply for the two coastal-upwelling stocks and documented as non-identified for jack mackerel. Third, we compute long-run comparative statics under a six-model CMIP6 ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR) for SSP2-4.5 and SSP5-8.5, reporting cross-model and within-posterior uncertainty separately. The variance decomposition (Appendices Online Appendix F and Online Appendix G) shows that, once the stock-dynamics steady-state collapse saturates the artisanal portfolio, the within-model component (posterior pooled with vessel heterogeneity) accounts for essentially the entire variance in projected fleet-level effort, so the policy-relevant uncertainty lies in the structural posterior rather than in disagreement across CMIP6 models.

The projected comparative statics indicate that the long-run intrinsic productivity of the coastal-upwelling pair—anchoveta and sardina común—declines sharply under all CMIP6 scenarios considered. Sardina común is consistently more exposed than anchoveta, with the SSP5-8.5 end-of-century scenario driving its steady-state productivity to floor levels at the Centro-Sur scale. The Centro-Sur jack mackerel stock, by contrast, does not provide sufficient information to identify a local climate elasticity in the 2000–2024 sample, consistent with its transboundary SPRFMO management. These results suggest that climate-adaptation policy in this fishery should target the quota-allocation architecture—specifically, the binding industrial–artisanal split and the limited cross-sector transferability that locks each fleet into its historical portfolio—rather than operational margins alone.

Several extensions to the present framework are natural. A first extension uses the posterior of $(r_i^0, K_i, \rho_i^{SST}, \rho_i^{CHL}, \Omega)$ identified here as a prior for a Stackelberg bi-level optimization, in which the regulator chooses the time path of TACs and the artisanal and industrial fleets allocate effort

across stocks under the climate-shifted law of motion. That exercise would deliver full forward biomass trajectories under CMIP6 anomaly paths with 90% credible bands, complementing the comparative-statics treatment in the present paper, and would follow the complete Kasperski (2015) approach with trip-level cost functions and an inverse-demand system. We leave it to future work; the posterior reported here is archived in machine-readable form to facilitate that extension. The spatial dimension of effort allocation is another natural extension, connecting the multi-species model to the location-choice literature (e.g., Dupont, 1993; Hicks et al., 2020; Smith, 2005), since the geographic distribution of species availability is likely to shift under warming. A direct route for this spatial extension is to migrate the annual NB trip equation to a daily discrete choice specification with environmentally informed species distribution models (SDMs) as proxies for local availability, following the approach of Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery. SDMs for the Chilean Humboldt Current System are an active area of development in IFOP and academic collaborations; once available at sufficient spatial resolution for anchoveta, sardina común, and jack mackerel, they would allow the present framework to capture daily and within-port heterogeneity in fishing-effort responses to climate-driven shifts in availability. Finally, incorporating risk preferences and production risk—through, for example, random-coefficient models—would provide a richer characterization of how fishers respond to the increased environmental variability projected under climate change (Kasperski & Holland, 2013).

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Data and code availability

All code required to replicate the analysis is openly available at <https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest> and will be archived at submission time as a versioned Zenodo release with a citable DOI. Environmental covariates are obtained from publicly available sources — the E.U. Copernicus Marine Service Information for the historical period (DOIs in references) and the CMIP6 archive accessed through the Earth System Grid Federation for the projection period. Stock-assessment biomass series are obtained from the Instituto de Fomento Pesquero (IFOP) annual technical reports for anchoveta and sardina común (Centro-Sur Chile) and from the South Pacific Regional Fisheries Management Organisation (SPRFMO) Scientific Committee reports for jack mackerel (transboundary). Annual catch (landings) series for 2000–2024 are obtained from the Servicio Nacional de Pesca y Acuicultura (SERNAPESCA) official desembarque database, covering all gears (industrial and artisanal), through a transparency request (AH010T0006857) filed under Chile’s Ley 20.285 in April 2025; aggregation and filtering to the Centro-Sur scale (regions V–X plus Ñuble) are reproducible from the source CSV via the script `R/01_data/99b_aggregate_catch_cs_from_sernapesca_v3.R`. The vessel-level trip and landings microdata used to estimate the negative binomial trip equation were obtained from IFOP under a Convenio Desempeño data-sharing agreement; these data are confidential at the vessel level and cannot be redistributed publicly, but are available to qualified researchers under equivalent agreements with IFOP. Vessel identifiers used in this paper have been anonymised before any code or output was archived. The replication repository contains aggregate-level processed data sufficient to reproduce all reported tables and figures.

7 Repository

The source code for this project is available at <https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest>.

8 References

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